Instituto Superior Técnico Master's Degree in Mechanical Engineering

Computational Fluid Mechanics

Homework 1 2026/2027 Group *10*

Student	E-mail	Number	Contribution
Martim Pestana	martim.pestana.fernandes@tecnico.ulisboa.pt	106138	25%
Francisco Ferreira	franciscobferreira@tecnico.ulisboa.pt	106428	25%
Pedro Pereira	pedro.nuno.sousa.pereira@tecnico.ulisboa.pt	106486	25%
David Ferreira	david.g.ferreira@tecnico.ulisboa.pt	107077	25%

Abstract

Finite-difference and cell-centred finite-volume discretisations were implemented and verified. The central-2 and backward-1 formulas exhibit observed orders of 2.00 and 1.00, and the steady convection problem converges with order 2.00. The linearised and transonic small-disturbance potential equations were solved over a diamond airfoil at $M_\infty = 0.3$, 0.91 and 2.2, with a measured Mach-wave angle of 27.07° against 27.04°. Newton's method with Mach continuation converged where Picard iteration diverged, and a clustered grid matched the accuracy of a uniform grid at half the cost.

1 Introduction

This work assesses the accuracy of finite-difference approximations and applies a cell-centred finite-volume method to small-perturbation potential flow over a diamond airfoil in the subsonic, transonic and supersonic regimes. The solutions are verified against analytical results and manufactured solutions, and the effects of the nonlinear solver and of the grid distribution are then explored.

2 Code

The code is written in Python 3.13 (NumPy, SciPy, Matplotlib). All functions are contained in the module `cfdlib`, and each exercise has a run script with only parameters and function calls (Table 1). Operators are assembled as sparse matrices, and every figure and table is reproduced without manual editing, as described in `README.txt`.

Table 1: Code workflow (outputs in `figures/`).

File	Role	Produces
`cfdlib/`	functions	–
`run_s1_accuracy.py`	Sec. 1.1	`s1_accuracy_*.pdf`
`run_s1_convection.py`	Sec. 1.2	`s1_convection_*.pdf`, `.txt`
`run_s2_flow.py`	Sec. 2	`s2_*.pdf`, `s2_*.txt`

3 Finite Differences

3.1 Formulas and truncation error

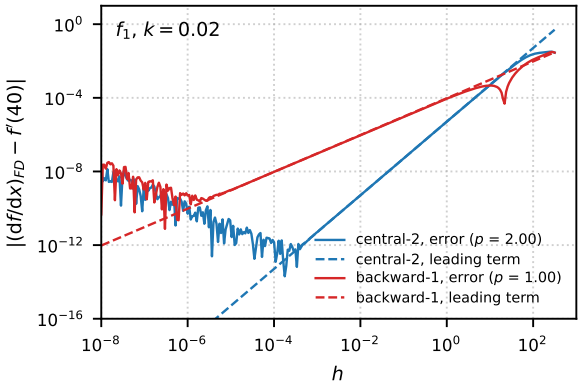
From Taylor expansions of $f_{i\pm1}$ about x_i, the backward-1 and central-2 formulas, with their leading truncation error, are

$$\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_i = \frac{f_i - f_{i-1}}{h} + \frac{h}{2} f_i'' + \mathcal{O}(h^2), \tag{1}$$

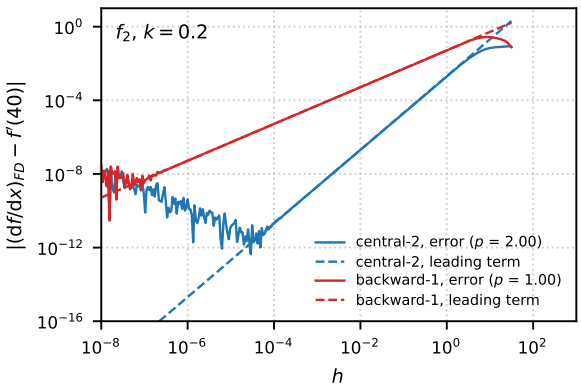
$$\left.\frac{\mathrm{d}f}{\mathrm{d}x}\right|_i = \frac{f_{i+1} - f_{i-1}}{2h} - \frac{h^2}{6} f_i''' + \mathcal{O}(h^4), \tag{2}$$

i.e. first- and second-order accurate, respectively.

3.2 Accuracy



(a) $f_1 = e^{\sin(0.02x)}$



(b) $f_2 = e^{\sin(0.2x)}$

Figure 1: Error of $\mathrm{d}f/\mathrm{d}x$ at $x = 40$ against h (solid) and leading truncation term (dashed), same axes. Slopes 1 and 2 in the asymptotic range; round-off below $h \sim 10^{-6}$–10^{-4}.

We computed the error at $x = 40$ for 300 values of h, from $h = 2\pi/k$ down to $h = 10^{-8}$ (Fig. 1). The dashed lines are the leading truncation terms of Eqs. (1) and (2), computed with the exact derivatives of f.

In the middle range of h the error and the dashed line are on top of each other, so the leading term describes

the error well. Fitting a line for $10^{-3} \le h \le 10^{-1}$ gives slopes of 2.000 (central-2) and 0.999 (backward-1) for f_1, and 2.000 and 1.001 for f_2. So the observed orders are 2 and 1, the same as the analytical ones: if h is halved, the backward-1 error is halved and the central-2 error drops by about 4.

For large h the curves move away from the dashed lines, because the higher-order terms of the Taylor series are no longer small. At $h = 2\pi/k$ the points $x - h$, x and $x + h$ have the same value of f, so both formulas give zero and the error is just $|f'(40)|$. For f_1, backward-1 has a dip near $h \approx 21$. This is not a better result: the error only changes sign there and passes through zero.

Comparing the two figures (same axes), the errors for f_2 are larger and the straight-line region starts at a smaller h. This is expected, because k is 10 times larger and the derivatives grow with k. What matters is how many points we have per wavelength, i.e. the value of kh.

For very small h the error grows again and becomes noisy. This is round-off error: we subtract two almost equal numbers and divide by a small h, so the error goes like ε/h ($\varepsilon \approx 10^{-16}$), with slope -1. The smallest error is around $h \approx 10^{-6}$ for backward-1 and $h \approx 10^{-4}$ for central-2. Central-2 reaches a smaller error ($\sim 10^{-12}$) than backward-1 ($\sim 10^{-10}$), and for the same h it is always more accurate.

3.3 Steady convection

The convection term of the steady-state problem was discretised with the second-order central difference mentioned in Section 3.2. (central-2) On a uniform grid with N + 1 points and spacing H = 40/N, the discrete equation at the interior nodes i = 1,. . . ,N-1 reads

$$a\,\frac{u_{i+1} - u_{i-1}}{2h} = s_i. \tag{3}$$

Equation (2) from HW1 is hyperbolic with a > 0, so information propagates in the +x direction: the solution is fixed at the inlet, x = 0, but at the last node (N = 40), we can't use central-2 for the sole reason of not having the N + 1 point, therefore, the second-order Backward difference

$$a\,\frac{3u_N - 4u_{N-1} + u_{N-2}}{2h} = s_N, \tag{4}$$

only requires upstream points, consistent with the direction of propagation while keeping overall second-order accuracy of the scheme.

Equations (3) and (4) are assembled into the linear system

$$a\,D\,u = s \tag{5}$$

where D is the sparse first-derivative operator (central rows in the interior, a backward-2 row at $x = 40$ and $\mathbf{s}$ is the source vector, evaluated from the exact solution $u = f_2(x)$ as

$$s(x) = a\,f_2'(x). \tag{6}$$

The inlet condition $u(0) = 1$ is enforced by replacing the first row of the system by $u_0 = 1$, and the system is then solved with a sparse direct (LU) solver.

For $N = 40$, the numerical solution is plotted against the exact solution $f_2(x)$, together with the pointwise error

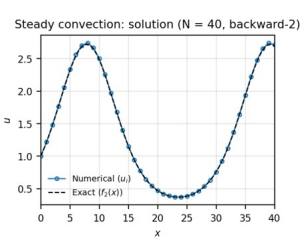
$$e_i = u_i - f_2(x_i). \tag{7}$$

The order of the method is then assessed with a grid-refinement study, halving h from $N = 10$ to $N = 2560$. For each grid, the error is measured with the discrete L_2 and L_∞ norms,

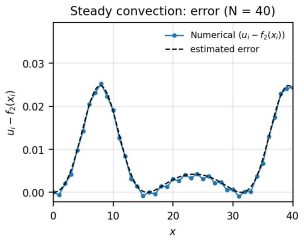
$$L_2 = \sqrt{h \sum_{i=0}^{N} e_i^2}, \tag{8}$$

$$L_\infty = \max_i |e_i|. \tag{9}$$

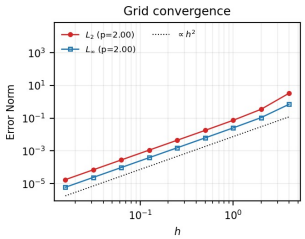
The factor h in (8) makes the L_2 norm an approximation of $\left(\int_0^{40} e^2\,dx\right)^{1/2}$, so it measures the error over the whole domain and can be compared between grids. The L_∞ norm gives the largest pointwise error. Both are plotted against h on logarithmic axes, where their slope gives the order. Since the leading truncation term of both (3) and (4) is proportional to h^2, both norms are expected to converge at second order. The order is fitted on the four finest grids, where the local orders have stabilised at 2.00; the coarsest grids (local orders 3.25 and 2.24) are not yet in the asymptotic range.



(a) Solution, $N = 40$



(b) Error $u_i - f_2(x_i)$, $N = 40$



(c) L_2 and L_∞ norms against h

Figure 2: Steady convection with central-2 inside the domain and backward-2 at $x = 40$.

4 Finite Volume

4.1 Discretisation

With $u = \partial\phi/\partial x$, both equations are written as a conservation law $\partial F(u)/\partial x + \partial^2\phi/\partial y^2 = 0$, with $F = (1 - M_\infty^2)u$ for (3a) and $F = (1 - M_\infty^2)u - \kappa u^2/2$,

$\kappa = (\gamma+1)M_\infty^2/U_\infty$, for (3b), so that $\mathrm{d}F/\mathrm{d}u = A$. Integrating over cell P (centroid x_i, faces w, e, s, n) and dividing by its volume,

$$\mu_i^c \frac{F_e - F_w}{\Delta x_i} + \mu_i^b \frac{F_w - F_{ww}}{\Delta x_{i-1}} + \frac{(\partial_y\phi)_n - (\partial_y\phi)_s}{\Delta y_j} = 0, \tag{10}$$

where face gradients are two-point differences between centroids. The two weights μ^c, μ^b are added in order to cover all the three regimes with one line. The central balance ($\mu^c = 1, \mu^b = 0$) is used for $M_\infty = 0.3$ and the balance shifted one cell upstream ($\mu^c = 0, \mu^b = 1$, first-order backward) for $M_\infty = 2.2$. At $M_\infty = 0.91$ the conservative Murman switch sets $m_i = 1$ where $A_i < 0$ and $\mu_i^c = 1 - m_i$, $\mu_i^b = m_{i-1}$: a shock point gets both balances ($F_e - F_{ww}$) and a sonic point neither, so the face fluxes telescope and the shock jump is conservative. The boundary conditions and solvers are in Table 2. The non-linear system $\mathbf{R}(\boldsymbol{\phi}) = 0$ is solved by Newton, $\boldsymbol{\phi} \leftarrow \boldsymbol{\phi} - \mathbf{J}^{-1}\mathbf{R}$, with $\mathbf{J} = \mathbf{X}\,\mathrm{diag}(A_f)\,\mathbf{G}_x + \mathbf{L}_{yy}$ (switch frozen in each step) and continuation $M_\infty = 0.80, 0.85, 0.88, 0.90, 0.91$; the linear equations converge in one step. Each linear system is solved by sparse LU.

4.2 Verification

The code was verified with manufactured solutions $\phi_m = X(x)Y(y)$ that satisfy the homogeneous conditions A–D of each regime, $X = \sin(\pi x/2L)$ (central, $M = 0.3$) or $X = \sin^2(\pi x/2L)$ (backward, $M = 2.2$), and $Y = \cos(\pi y/H) + (1 - y/H)^3$, whose non-zero $\partial_y\phi_m(x, 0)$ is imposed as the wall flux E. The source $f = (1 - M^2)\phi_{m,xx} + \phi_{m,yy}$ is evaluated at the centroids, and errors are measured against ϕ_m at the centroids, separately in interior cells and in the cells that touch a boundary. Boundary values enter as face values: Dirichlet over the half-cell distance $\Delta x/2$, Neumann as an imposed face flux. Both schemes reach their design order, also at the boundaries (Fig. 4 and Table 3): the $\mathcal{O}(h)$ local error of the half-cell Dirichlet gradient is confined to one layer of cells and does not degrade the global second order.

4.3 Results

The local state follows from Eq. (4) of the statement with u, v from the face gradients (cell average in central cells, upwind face in backward cells). Figure 3 shows the three regimes.

Subsonic. The disturbance is elliptic and fore–aft symmetric: ϕ is antisymmetric about mid-chord and the perturbations decay in all directions. The flow is practically incompressible ($0.284 \le M \le 0.341$, $0.987 \le \rho/\rho_\infty \le 1.005$); the wall C_p ranges from $+0.12$ at the leading and trailing edges to -0.30 at the shoulder, where linear theory has a logarithmic singularity. The drag integral $C_d = 2\int C_p\, y_s'\, \mathrm{d}x/c$ is zero to five digits, as d'Alembert's paradox requires.

Transonic. The flow expands around the shoulder and becomes sonic exactly there, the classical TSD result for a convex corner; a supersonic pocket ($M_{\max} = 1.34$, 665 cells) extends $2.2c$ above the wall and is closed by a shock that reaches the wall at $x \approx 4.03c$, just behind the trailing edge. The compression and expansion are no longer symmetric: the shock produces wave drag, $C_d = 0.0409$, and the density drops to $0.68\rho_\infty$ in the pocket. Newton converges quadratically in 7–10 steps per Mach level (43 steps in total, 4.2 s).

Supersonic. The disturbances are confined to the Mach waves from the leading edge, mid-chord and trailing edge, and the flow ahead of the leading-edge wave is undisturbed. The wall $C_p = \pm 0.062$ matches Ackeret's $\pm 2\theta/\sqrt{M_\infty^2 - 1} = \pm 0.0612$, and $C_d = 0.00687$ is 6.5% below Ackeret's 0.00735 because the first-order scheme smears the pressure jumps at the corners. The Mach-wave angle, measured from the steepest compression in each row ($0.05c \le y \le 0.5c$), is 27.07° against $\arcsin(1/2.2) = 27.04°$. Farther from the wall the smeared wave biases a threshold detector (Table 4); both converge to the theoretical angle with refinement.

4.4 Exploration

4.4.1 Treatment of the non-linearity: Newton vs Picard

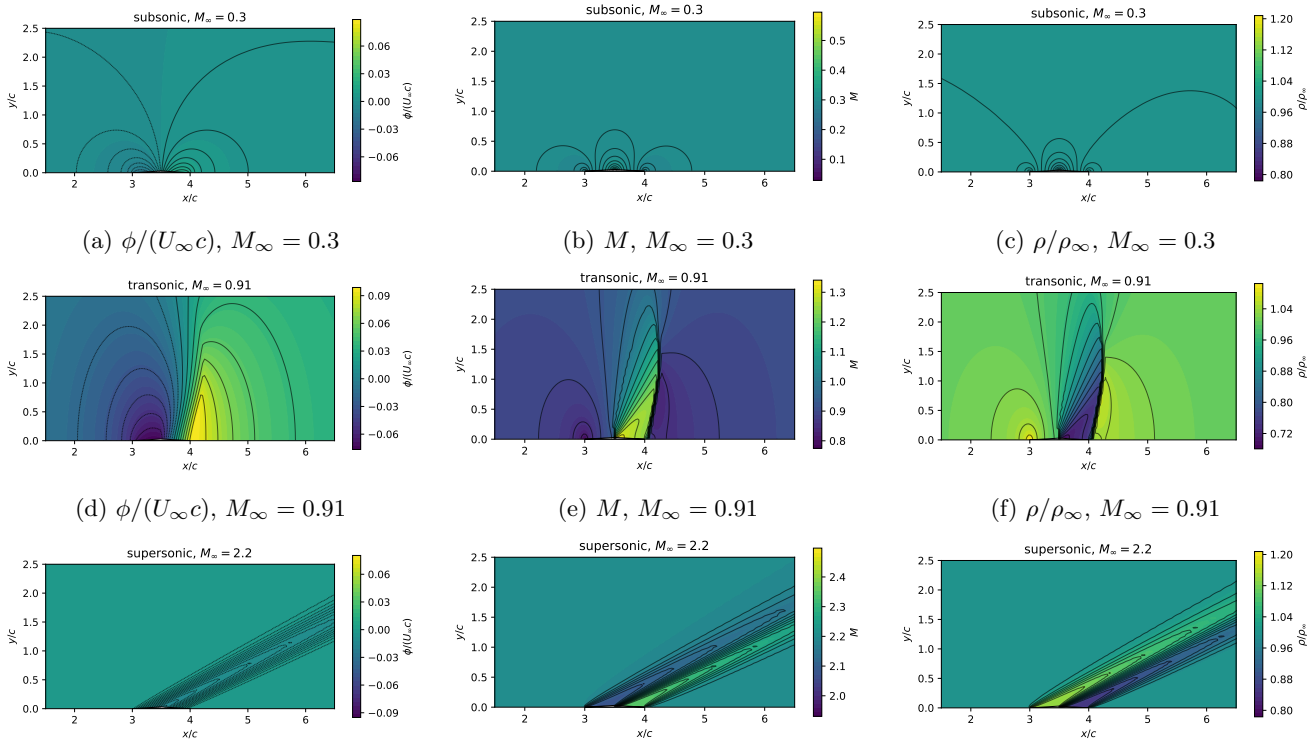
The first way we chose to explore the problem was trhough the comparison between Newton vs Picard. We replaced Newton by Picard iteration, which freezes the secant coefficient $a = F(u)/u = (1 - M_\infty^2) - \kappa u/2$ and the switch, solves $\mathbf{X}\,\mathrm{diag}(a)\mathbf{G}_x\boldsymbol{\phi}^* + \mathbf{L}_{yy}\boldsymbol{\phi}^* = \mathbf{b}$ and under-relaxes $\boldsymbol{\phi} \leftarrow \boldsymbol{\phi} + \omega(\boldsymbol{\phi}^* - \boldsymbol{\phi})$; its fixed point is the same discrete solution. Picard diverged for every ω tried, with and without continuation, and converged only at $M = 0.80$, where the pocket has 16 cells (Table 5 and Fig. 6). The reason is a type mismatch: the switch upwinds where the tangent $A = (1 - M_\infty^2) - \kappa u < 0$, but the secant $a = \frac{1}{2}\left[(1 - M_\infty^2) + A\right]$ stays positive while $-(1 - M_\infty^2) < A < 0$, i.e. for perturbations between the sonic value and twice it, so Picard freezes an elliptic-signed coefficient inside a backward (hyperbolic) balance—an anti-diffusive operator. In the converged solution, 221 of the 702 faces used by backward balances are in that state. Newton linearises with A itself and stays type-consistent, but it also needs continuation: started from $\phi = 0$ at $M = 0.91$ it diverges, because the first linearisation (around the linear solution, with a large spurious pocket) is too far from the root.

4.4.2 Uniform vs non-uniform meshes: accuracy against cost

We compared three mesh families on the same domain (Fig. 7): U (uniform in x and y), G (uniform in x, geometric in y; the mesh of Fig. 3) and C (uniform cells over $[x_{LE} - 0.5c,\, x_{TE} + 0.5c]$, geometric growth 1.05 outside, geometric in y), each refined by factors 0.5, 1 and 2 in both directions. The error in C_d and in the wall C_p on the chord is measured against the finest C mesh (360×360), which needed a longer continuation (from $M = 0.5$ in steps of 0.01). Figure 5 and Table 6 show that (i) stretching in y is what buys accuracy: for the same 21 600 cells, U has five times the C_d error of

Table 2: Discretisation, boundary conditions and solver per regime ($y'_s = \pm 0.06$; face values of y'_s are cell means, exact for the diamond).

Regime	x-fluxes	y-fluxes	A (inflow)	B (far field)	C (outflow)	D (symmetry)	E (airfoil)	Solver
Subsonic	central	central	$\phi = 0$	$\phi_y = 0$	$\phi_x = 0$	$\phi_y = 0$	$\phi_y = U_\infty y'_s$	LU
Transonic	Murman switch on A_i	central	$\phi = 0$	$\phi_y = 0$	$\phi_x = 0$	$\phi_y = 0$	$\phi_y = U_\infty y'_s$	Newton +
Supersonic	backward (1st order)	central	$\phi = \phi_x = 0$	$\phi_y = 0$	none	$\phi_y = 0$	$\phi_y = U_\infty y'_s$	LU



(a) $\phi/(U_\infty c)$, $M_\infty = 0.3$ (b) M, $M_\infty = 0.3$ (c) ρ/ρ_∞, $M_\infty = 0.3$

(d) $\phi/(U_\infty c)$, $M_\infty = 0.91$ (e) M, $M_\infty = 0.91$ (f) ρ/ρ_∞, $M_\infty = 0.91$

(g) $\phi/(U_\infty c)$, $M_\infty = 2.2$ (h) M, $M_\infty = 2.2$ (i) ρ/ρ_∞, $M_\infty = 2.2$

Figure 3: Fields of $\phi/(U_\infty c)$, M and ρ/ρ_∞ for the three regimes, grid 240×90 (uniform $\Delta x = c/30$, geometric Δy from $0.02c$, ratio 1.043), domain $8c \times 20c$, leading edge at $x = 3c$. Each field uses the same colour-range width in the three regimes; the thin lines are isolines.

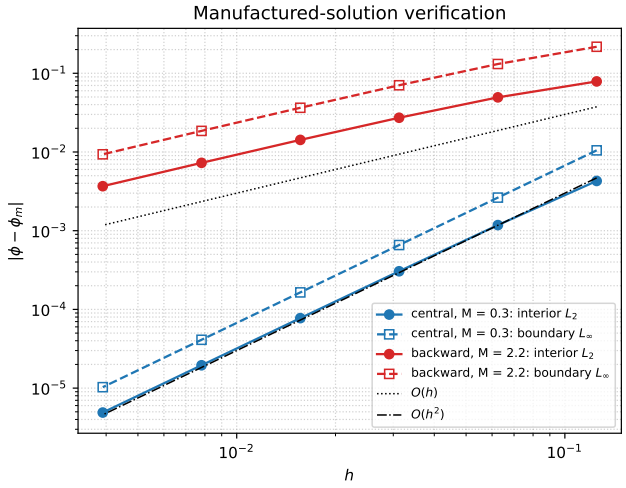


Figure 4: Manufactured-solution error against h on uniform $N \times N$ grids: the central scheme follows h^2 and the backward scheme h, in the interior and in the boundary cells.

Table 3: Observed orders of accuracy (between consecutive grids).

	central, $M = 0.3$		backward, $M = 2.2$	
$N_x \times N_y$	p (int.)	p (bnd.)	p (int.)	p (bnd.)
8^2	–	–	–	–
32^2	1.95	2.00	0.86	0.90
64^2	1.98	2.00	0.94	0.95
128^2	1.99	2.00	0.97	0.98
256^2	2.00	2.00	0.98	0.99

Table 4: Mach-wave angle (deg) against refinement; $\arcsin(1/M_\infty) = 27.04°$.

Grid	threshold ($0.3 \le y/c \le 1.5$)	steepest ($0.05 \le y/c \le 0.5$)
240×90	28.43	27.07
480×180	27.32	26.93
960×360	27.04	26.98

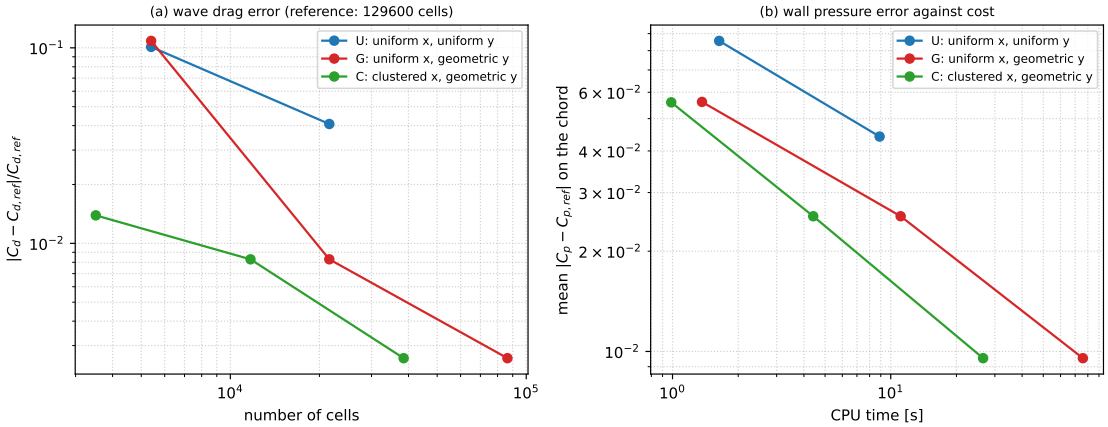


Figure 5: Mesh families at $M_\infty = 0.91$: (a) relative C_d error against the number of cells; (b) mean wall-C_p error on the chord against CPU time. Reference: C mesh 360×360, $C_d = 0.04051$.

Table 5: Non-linear solvers at $M_\infty = 0.91$ (240×90, tolerance 10^{-10}). Iterations count the residual evaluations over all continuation levels (Newton: 43 steps).

Method	Iter.	CPU [s]	Outcome
Newton, continuation	48	4.2	converged
Newton, no continuation	16	2.3	diverged
Picard $\omega = 1.0$, continuation	319	30.5	diverged
Picard $\omega = 0.5$, continuation	318	31.4	diverged
Picard $\omega = 0.2$, continuation	275	26.4	only $M = 0.80$
Picard $\omega = 0.1$, no continuation	32	3.2	diverged

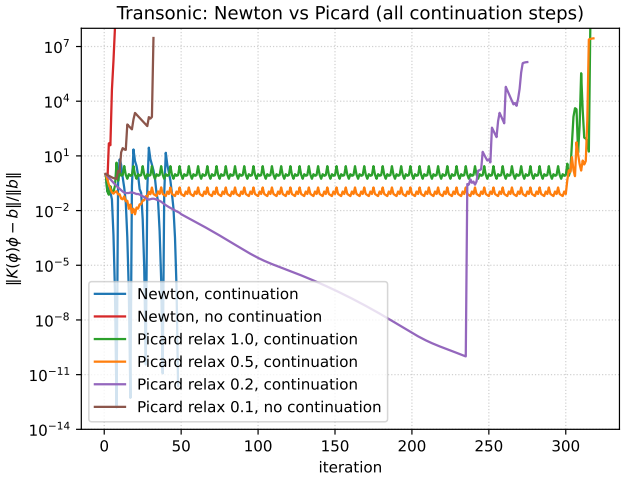


Figure 6: Residual histories over all continuation levels. Newton: quadratic convergence at each level; Picard: converges at $M = 0.80$ with $\omega = 0.2$ only, otherwise stalls or diverges.

G, since its wall cell is eleven times thicker; (ii) clustering in x costs nothing: C gives exactly the errors of G with 46–54% of the cells and 35–40% of the CPU time, the removed cells lying far upstream and downstream; (iii) at the coarsest level C is far better (1.4% vs 10.9%) because its faces fall on the shoulder, where the slope jumps, while those of G do not; (iv) the observed order is about 1.7 for C_d and 1.4 for C_p, limited by the shock and the first-order upwinding; (v) Newton did not converge on the finest uniform mesh, so the mesh also affects robustness.

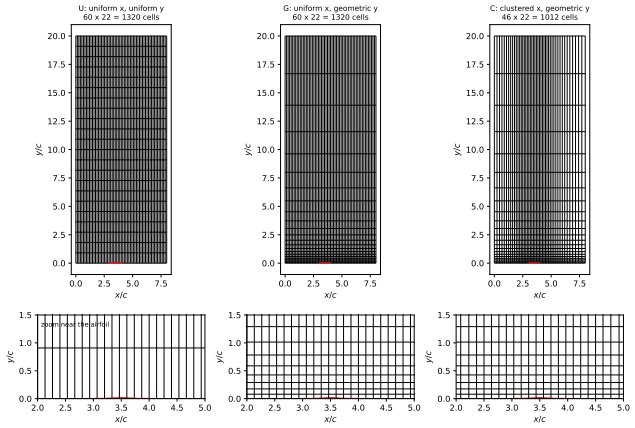


Figure 7: The three mesh families, drawn at a quarter of the base resolution (top: whole domain; bottom: near the airfoil, in red).

Table 6: Mesh families at $M_\infty = 0.91$ (errors relative to the reference).

Mesh	Cells	CPU [s]	$\Delta C_d/C_d$	C_p err.
U $\times 1$	21 600	8.9	4.1%	0.044
G $\times 1$	21 600	11.1	0.83%	0.026
C $\times 1$	11 700	4.4	0.83%	0.026
U $\times 2$	86 400	68.9	not converged	
G $\times 2$	86 400	76.0	0.26%	0.0096
C $\times 2$	38 520	26.5	0.26%	0.0096

5 Conclusions

The finite-difference study confirmed the expected orders of accuracy: the central-2 and backward-1 formulas showed slopes of 2.00 and 1.00 for both functions, as long as the step was small enough for the leading truncation term to dominate and large enough to avoid round-off. The error depends mainly on the number of points per wavelength, so f_2 needs a step about ten times smaller than f_1 for the same error. In the steady convection problem, closing the central-2 scheme with a backward-2 formula at the outflow kept the second order of the whole solution, with observed orders of 2.00 in both norms.

The finite-volume code reached order 2 with central differences and order 1 with backward differences, in the interior and at the boundaries. The subsonic and supersonic solutions agree with linear theory, giving zero drag at $M_\infty = 0.3$ and, at $M_\infty = 2.2$, the Ackeret wall pressure and a Mach-wave angle of 27.07° against 27.04°. At $M_\infty = 0.91$ the Murman switch captured a supersonic pocket closed by a shock just behind the trailing edge, with $C_d = 0.0409$. This case only converged with Newton's method and continuation in Mach number, while every Picard variant tested diverged. Finally, refining the mesh near the wall was what improved the accuracy, and concentrating cells around the airfoil in x gave the same result with about half the cells and 40% of the computing time.

References

[1] E. M. Murman and J. D. Cole, Calculation of plane steady transonic flows, *AIAA Journal* 9(1):114–121, 1971.

[2] E. M. Murman, Analysis of embedded shock waves calculated by relaxation methods, *AIAA Journal* 12(5):626–633, 1974.

[3] J. D. Anderson, *Modern Compressible Flow*, 3rd ed., McGraw-Hill, 2003.

[4] J. H. Ferziger, M. Perić and R. L. Street, *Computational Methods for Fluid Dynamics*, 4th ed., Springer, 2020.

[5] P. J. Roache, *Verification and Validation in Computational Science and Engineering*, Hermosa, 1998.

[6] P. J. Roache, *Verification and Validation in Computational Science and Engineering*, Hermosa, 1998.